

Sampled evaluation — $t = 2016$ (5:1 negative ratio, same protocol)

Method	PR-AUC	AUROC	F1	NDCG@20	Prec@20	CWR
RCA Persistence	0.529	0.653	0.441	0.519	0.492	0.335
Density	0.366	0.782	0.428	0.473	0.431	0.854
ECI	0.137	0.468	0.261	—	—	0.432
KNN (LLM embeddings)	0.246	0.641	0.316	0.342	0.320	0.655
XGBoost (unoptimized)	0.643	0.895	0.629	0.683	0.619	0.932
XGBoost (optimized)	0.662	0.901	0.638	0.698	0.634	0.938
GNN-4F	0.424	0.819	0.473	0.430	0.392	0.879
GNN-11F	0.433	0.823	0.479	0.452	0.412	0.883
GNN-11F+LLM	0.443	0.829	0.485	0.467	0.421	0.891
GNN-11F-LLM + RCA-skip (unoptimized)	0.639	0.893	0.633	0.675	0.616	0.927
GNN-11F-LLM + RCA-skip (optimized)	0.640	0.893	0.630	0.660	0.606	0.930

Bold row = best method overall (wins the most metrics: XGBoost (optimized)). ECI has no within-country ranking signal, so NDCG@20 / Prec@20 are marked — rather than reported off an arbitrary tie-break.